

✓ CHAPTER II – REVIEW OF RELATED LITERATURE AND STUDIES

📌 PURPOSE OF THIS CHAPTER

This chapter provides the **intellectual foundation** of your study. It shows that your project:

- Is **grounded in scholarly work** and prior system development.
 - Builds on **existing knowledge and systems**, not just opinions.
 - Is **unique**, while being inspired by proven approaches.
-

■ STRUCTURE OVERVIEW:

Section	Purpose
A. Related Literature	Presents academic theories, concepts, and frameworks from books and journals.
B. Related Studies	Presents similar projects (local/foreign), their features, and results.
C. Table of Comparison	Visually compares the previous studies with your own system.

◆ A. RELATED LITERATURE

📖 What is it?

These are **theoretical or academic sources** from books, journals, or scholarly articles. This section should explain:

- Concepts like **Inventory Management, Database Systems, ISO 25010**, etc.
- Why these concepts are relevant to your study.

Steps to Write:

1. Choose **3–5 core concepts** related to your system.
Example: Inventory Systems, Database Design, ISO Software Evaluation.
2. Look for **academic sources** (books, journals, e-books, PDFs).
3. Paraphrase and summarize their explanation.
4. Use proper **APA 7th edition citations**.
5. After each summary, write a sentence that **connects** it to your system.

Full Example:

Inventory Management and Its Importance

Silver, Pyke, and Thomas (2016) emphasized that inventory management systems are essential in controlling stock levels, minimizing human error, and preventing tool loss. They explained that effective inventory systems help institutions avoid disruptions in operations by ensuring that required items are always available.

This literature supports the need for a system in the automotive laboratory to prevent delays during hands-on exercises caused by missing tools.

Database Systems in Inventory Applications

Coronel and Morris (2016) described a database as an organized collection of structured information that is accessed and managed through a Database Management System (DBMS). They discussed how databases ensure data consistency, support multiple users, and allow secure retrieval of records.

In the context of the proposed Tools Inventory System, the database will serve as the central storage of all transaction data such as stock-in/out, borrowing, and tool availability.

Software Evaluation Standards – ISO 25010

Pressman (2010) introduced the ISO 25010 model, which defines eight characteristics to evaluate software quality: functionality, usability, reliability, maintainability, efficiency, portability, compatibility, and security.

This study uses the ISO 25010 standard to evaluate the Tools Inventory System after deployment to ensure its reliability and usability.

◆ B. RELATED STUDIES

What is it?

This part reviews **actual system development projects** similar to yours. These may be:

- **Undergraduate/Capstone systems**
- **Graduate theses**
- **Published system research**

They can be **local or foreign** studies. This section shows that:

- Your idea is **relevant** and has been attempted before.
 - Your system **has something new** to contribute.
-

Steps to Write:

1. Identify **3–5 existing projects or studies** with similar goals.
2. For each:
 - State the **author(s)** and **year**
 - Summarize the **purpose**
 - Summarize the **features** or **results**
 - Optionally say **how it's different from or similar to your system**
3. Use APA citation for the authors.

✓ **Full Example:**

Development of a Laboratory Tools Tracking System

Delos Reyes (2018) developed a tool-tracking system for a state university's mechanical lab, aiming to address frequent tool losses and incomplete records. The system automated borrowing and returning records, reducing manual errors and improving student accountability.

This is similar to the proposed study, but it lacked a formal system evaluation like ISO 25010.

Web-Based Inventory System for Technical Colleges

Chen (2020) created a web-based inventory system for a technical college's engineering department. It allowed remote access to tool records and improved transaction logs and report generation.

Unlike this study, the Tools Inventory System will be LAN-based, limiting access to the laboratory area only.

RFID-Enhanced Tool Inventory Management

Garcia and Liu (2019) implemented an inventory system that used RFID tags to track tools in real time. The system provided alerts for overdue tools and generated analytics on tool usage patterns.

While RFID improves tracking, this feature is not included in the current system due to cost and technical constraints. However, it is considered a future enhancement.

◆ C. TABLE OF COMPARISON

What is it?

This table **summarizes the differences and similarities** between:

- The **related studies**
- Your **current study**

It helps readers **see at a glance** what features are covered by each system and what makes your system **unique**.

Steps to Create:

1. List **key system features** on the left column.
 2. List the **authors/studies** across the top.
 3. Add a final column for **“This Study”** (your project).
 4. Use checkmarks () or  to indicate presence/absence of each feature.
-

Full Example Table:

Features	Delos Reyes (2018)	Chen (2020)	Garcia & Liu (2019)
Tool Registry			
Borrow/Return Transactions			
Inventory Report Generation			
RFID Integration			
Web-Based Access			
ISO 25010 System Evaluation			
Student Access via Login			

REFERENCES

- Chen, R. (2020). Web-Based Inventory System for Technical Colleges. **Journal of Computing Education**, 12(2), 43–56.
- Coronel, C., & Morris, S. (2016). **Database systems: Design, implementation, & management** (12th ed.). Cengage Learning.
- Delos Reyes, A. (2018). Development of a Laboratory Tools Tracking System. Unpublished undergraduate thesis, Polytechnic University of the Philippines.
- Garcia, R., & Liu, T. (2019). RFID-Enhanced Inventory System in Automotive Labs. **International Journal of Information Technology**, 10(1), 66–72.
- Pressman, R. (2010). **Software engineering: A practitioner's approach** (7th ed.). McGraw-Hill Education.
- Silver, E. A., Pyke, D. F., & Thomas, D. J. (2016). **Inventory and production management in supply chains** (4th ed.). CRC Press.